Yu Li

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Education

George Washington University — **Washington, D.C.**
Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0 — *2025.08 – 2029.05 (expected)*
Advisor: Prof. Tian Lan; Mentor: Prof. Zhengling Qi

Wuhan University, Hongyi Honor College — **Wuhan, China**
B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0 — *2021.09 – 2025.06*

Industry Experience

ByteDance Inc. — **San Jose, CA**
Machine Learning Scientist Intern, Tiktok Recommendation Core — *2026.05 – 2026.08*
Worked on production-scale video recommendation at TikTok, building LLM-based ranking models for the ranking stage and integrating content signals into ranking features.

Research Highlights

- **LLM Reasoning:** Developing credit assignment, preference optimization, and context conditioning methods to improve alignment and reasoning in LLMs [P1][P3][P4][P5].
- **Agentic RL:** Designing reinforcement learning frameworks for hierarchical skill acquisition and multi-turn policy optimization across single and multi-agent systems [P2][C2][C5].
- **Efficient AI:** Improving training and inference efficiency for LLMs through parameter-efficient fine-tuning and long-context KV cache [C1][C6][C7].
- **Generative Models:** Exploring diffusion-based generative models for personalized content-style synthesis and cross-modal generation [C3][J2].

Preprints & Under Review

P=Preprint

[P1] OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning
Yu Li, Rui Miao, Tian Lan, Zhengling Qi

[P2] ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning
Yu Li, Rui Miao, Zhengling Qi, Tian Lan

[P3] Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO
Yu Li, Tian Lan, Zhengling Qi

[P4] INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization
Yu Li, Tian Lan, Zhengling Qi
AI with Recursive Self-Improvement Workshop at ICLR, 2026

[P5] Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning
Yu Li, Shu Hong, Tian Lan
Under review at EMNLP 2026, OA: 3.33/5

Publications

C=Conference, J=Journal, †=Equal Contribution

I have published **5** papers in top-tier machine learning and computer vision venues (e.g., COLM, ACL, CVPR, AAAI, ICASSP), with **4** of them as **first author**.

[C1] MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference
COLM, 2026
Yu Li, Binxu Li, Tian Lan

[C2] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization
ACL Findings, 2026
Yu Li, Sizhe Tang, Tian Lan

[C3] CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion
CVPR, 2026
Yu Li, Yujun Cai, Chi Zhang

[C4] KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models
ICASSP (Oral), 2026
Yu Li, Da Chang, Xi Xiao

[C5] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense
INFOCOM Intelligent Cloud Computing and Networking, 2026
Yu Li†, Sizhe Tang†, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

[C6] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT
AAAI, 2026
Da Chang, Peng Xue, **Yu Li**, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

[C7] Mixed Text Recognition with Efficient Parameter Fine-Tuning and Transformer
ICONIP, 2024
Yu Li†, Da Chang†

[J1] Dual branch SAM-Transformer Fusion Network for Accurate Breast Ultrasound Image Segmentation
Medical Physics, 2025
Yu Li, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

[J2] SfMDiffusion: Self-supervised Monocular Depth Estimation in Endoscopy Based on Diffusion Models
International Journal of Computer Assisted Radiology and Surgery, 2025
Yu Li, Da Chang, Jin Huang, Du Wang, Liye Mei, Cheng Lei *et al.*

Selected Awards & Honors

- **ICML 2026 Gold Reviewer**, top reviewers recognition. *2026*
- **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. *2024*
- **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. *2023, 2024*
- **First-Class Scholarship**, Top 5%, ¥3,000, three times. *2022, 2023, 2024*

Academic Services

- **Conference Reviewer:** {NeurIPS, COLM, ACM MM, ICML, ICLR, AAAI, ICASSP}'2026
- **Journal Reviewer:** TMLR, TPAMI, ToN, Neurocomputing